Virginia Polytechnic Institute and State University

Grado Department of Industrial & Systems Engineering

ISE 5405 – Optimization I

Fall 2026

Course Overview

Catalog Description

(3 units) Introduction to theory of linear programming. A mix of theoretical concepts and numerical algorithms to solve the linear programming problems. Modeling for real world problems using linear programming and integer linear programming. Geometric foundations for linear programs – characterization for polyhedral sets and convex analysis. Numerical algorithms for linear programs: simplex method (its geometry and algebra), primal-dual simplex algorithm, revised simplex, two-phase and big-M methods. Farkas' Lemma and Optimality Karush-Kuhn-Tucker (KKT) conditions for linear programs. Duality theory, sensitivity analysis, state-of-the-art modeling language and solvers.

Prerequisites

There is no formal prerequisite, but a good knowledge of linear algebra and calculus, along with strong background in mathematics and basic coding skills, are required. Having an introductory OR class is a plus and helpful.

Pre-course Preparation

Undergraduate course: Linear Algebra - MIT OpenCourseWare (Lectures 1-10), by Gilbert Strang (available online).

Course Logistics and Instructional Team

Meetings and Canvas

Meeting time: Tuesdays and Thursdays, 9:30–10:45 a.m.

Meeting place: Durham Hall 261

Course website: ISE 5405 on Canvas

Instructor

Name: Dr. Jiaxiang "Jason" Li

Office: Durham Hall 213

Homepage: jasonjiaxiangli.github.io

E-mail: jasonljx@vt.edu

Office hours: Thursdays, 11:00 a.m.–12:00 p.m.

Zoom meeting ID: `89033654462` (for office hours)

(Prof. Robert Hildebrand will teach the August 25 and August 27 class meetings online)

Teaching Assistant

Name: Asha Barua

E-mail: ashabarua@vt.edu

TA office hours: Tuesdays, 11:30 a.m.–12:30 p.m.

Location: Durham Hall 214

Zoom meeting: Available upon request; meeting ID: `ashabarua`

Course Learning Goals

Course Objectives

This is a course on theory of linear programming. It is rigorous, covers a large amount of material on optimization foundational theory and algorithms, and is meant to prepare students for research in optimization. Students will be expected to read the textbook and be prepared to engage in in-class practice and discussion. In this course, we will be mainly studying linear optimization techniques for modeling and solution of deterministic linear models. Linear programming is a basic optimization method that is used as a building block to many other optimization algorithms and software. Therefore, solid knowledge of linear programming is essential for those who study optimization. An important aspect of the course involves the modeling and solution implementation of a mathematical optimization problem using the optimization software Gurobi.

Student Learning Objectives

1. Develop mathematical models for real-world problems using linear programs.

2. Build geometric foundations for linear programs, in particular characterize polyhedral sets and learn convex analysis.
3. Apply simplex, revised simplex, two-phase, big-M, and primal-dual simplex methods for solving linear programming problems along with analyzing characteristics and implementation issues associated with numerical algorithms.
4. Derive dual and KKT optimality conditions for linear programs.
5. Interpret the outputs from linear programming models and evaluate their sensitivity to changes in input.
6. Utilize state-of-the-art modeling languages and optimization solvers to solve linear programs.
7. Communicate the formulation, analysis, and results of an optimization model clearly and professionally in written and/or oral form.

Course Materials and Software

Canvas Communication

I will use Canvas to post readings, homework assignments and their solutions, and other information about the course. I will use Canvas announcements, which you will also receive by e-mail, to communicate regularly with the class. Please check Canvas regularly and keep up with the announcements; otherwise, you may miss important information.

Textbook

- D. Bertsimas and J. N. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, Belmont, Massachusetts, 1997.

Other References

- R. Hildebrand, L. Poirrier, D. Bish, D. Moran. *Mathematical programming and operations research*. https://open-optimization.github.io/open-optimization-or-book/
- M. S. Bazaraa, J. J. Jarvis, H. D. Sherali, *Linear Programming and Network Flows*, 4th ed., John Wiley & Sons, Inc., New York, NY 2010. (E-book is available from VT library's website).
- G. B. Dantzig, M. N. Thapa, *Linear Programming 1: Introduction*, Springer, 1997.
- G. B. Dantzig, M. N. Thapa, *Linear Programming 2: Theory and Extensions*, Springer, 2003.
- D. G. Luenberger, Y. Ye, *Linear and Nonlinear Programming*, Springer, 3rd ed., Springer, 2008. (Part I of this book is relevant for this class).

Gurobi and Python

We will use **Gurobi** with its `gurobipy` Python interface for solving linear optimization models in this course. Gurobi is a state-of-the-art optimization solver that is widely used in both industry and academia. Students should install the free academic version of Gurobi from https://www.gurobi.com/, which includes access to the Python interface `gurobipy`.

Assignments will require you to write optimization models in Python and solve them using `gurobipy`. A brief tutorial on Gurobi and Python modeling will be provided in class. Familiarity with Python is strongly recommended. If time permits, we may also explore other solvers or modeling tools. All solver-related questions should be directed to the class TA.

Assessment and Grading

Homework and Canvas Submissions

There will be 5 graded homework assignments and one non-graded assignment, approximately one every two weeks. The five graded homework assignments are weighted equally within the 50% homework category, so each graded homework contributes 10% of the final course grade. The non-graded assignment does not affect the course grade. You are expected to attempt each question on your own before seeking help. You may discuss homework problems (except for any designated bonus assignments) with your classmates; however, you must write up and submit your own solutions independently. Plagiarism—copying someone else's work—will not be tolerated and will be treated as a violation of the Graduate Honor Code.

Each homework submission attempt must include one complete PDF file containing all written solutions. If the assignment involves coding, please also upload any supporting code files (e.g., `.py` or `.ipynb` files) in the same Canvas attempt. Make sure your code is well-organized and clearly referenced in your written solutions.

Canvas Submission Policy. Students may make an unlimited number of Canvas submission attempts before the assignment's posted cutoff date and time, including during the four-day late-acceptance window. Ordinarily, only the most recent complete attempt received by the cutoff will be graded, and any applicable late penalty will be determined by the Canvas timestamp of that attempt; therefore, a replacement submitted after the due date is treated as late even if an earlier attempt was submitted on time. If a student makes an accidental replacement submission or experiences a documented technical problem, the student may request within 24 hours that a specific earlier Canvas attempt be graded instead. Such requests are granted only at the instructor's discretion and must refer to an attempt already submitted through Canvas; they may not be used to submit additional or revised work by email. Students are responsible for allowing sufficient time for uploads to complete. Internet, browser, or upload delays near the deadline do not alter the Canvas timestamp used to determine lateness; exceptions may be made for documented Canvas outages or other extraordinary technical failures. **Absent an approved exception, the most recent complete Canvas attempt will be graded.** Students are responsible for verifying that the intended PDF and all required supporting files are attached and readable. Canvas will not accept submissions after the posted cutoff.

Late Work

I encourage you to submit all homework by the specified due date and time. Late homework will be accepted until the cutoff shown in Canvas, four calendar days (96 hours) after the due date and time, with the penalties shown below. Any fraction of a 24-hour period after the deadline counts as a full day late. As stated above, the penalty is determined from the timestamp of the most recent submission attempt that will be graded.

Assignment Late Penalty

Days Late	Late Penalty %
1	15%
2	30%
3	45%
4	60%

Exams

There will be a midterm on **Thursday, October 15** during the regular class time in the regular classroom, Durham Hall 261, and a comprehensive final on **Monday, December 14, 2026, from 7:45–9:45 a.m.** The final exam will be held in the same classroom where the course regularly meets, Durham Hall 261. Both exams are closed book and closed notes. These dates follow the official university exam schedule. Except in cases of university-authorized absences, documented emergencies, or approved accommodations, no make-up exams will be offered. Students who believe they qualify for an exception should contact the instructor as soon as possible and provide any required documentation.

In-class Practice and Discussion (Ungraded)

Class meetings will include short problem-solving exercises completed individually and with classmates. These activities are intended solely for practice, immediate feedback, and discussion of solution strategies with peers; they will not be collected, scored, or included in the calculation of the final course grade. Attendance and active engagement are strongly encouraged because they support learning, but they are not graded. Students should bring a laptop and charger when requested so they can take part in computational or modeling exercises during class.

Grading Calculation

The grades will be calculated based on the following grading schema:

When preparing your assignments and exam solutions, pay attention to their content, cleanliness, and organization; they all contribute to your grade. Final course percentages are calculated from unrounded component scores using the stated category weights, then rounded to two decimal places before the letter-grade cutoffs are applied. Any course-wide curve will be applied uniformly and may raise, but will not lower, a student's letter grade.

Grading Breakdown

Category	Percentage
Homework	50%
Midterm Exam	25%
Final Exam	25%
Total	**100%**

Letter Grade Scale

Range	Grade
93.00–100.00	A
90.00–92.99	A-
87.00–89.99	B+
83.00–86.99	B
80.00–82.99	B-
77.00–79.99	C+
73.00–76.99	C
70.00–72.99	C-
67.00–69.99	D+
63.00–66.99	D
60.00–62.99	D-
Below 60.00	F

Regrade Requests

Submit all regrade requests to the instructor, **not the course teaching assistant (TA)**, within one week (seven calendar days) from the date the scores are posted; a Canvas announcement will be sent when scores are available.

- **Homework assignments:** E-mail the instructor with a specific explanation of the suspected grading error, referring to the posted solution key.
- **Exams:** On a separate sheet of paper, explain each suspected grading error by referring to the posted solution key. Staple that sheet to the front of the exam and submit both items directly to the instructor.

A regrade request is not an opportunity to revise or add to the original work. Other portions of the homework assignment or exam may also be regraded as deemed appropriate. The regraded score will replace the original score regardless of whether it is higher than, the same as, or lower than the original score. Regrade requests submitted after the one-week deadline will not be considered.

Course Schedule and Major Deadlines

Below are the planned due dates for homework and exams. Homework will be due approximately every two weeks on Thursdays. Homework due dates are subject to change; the final-exam date and time are set by the official university exam schedule.

- **Aug 25 (Tue):** First class meeting
- **Sep 10 (Thu): Homework 1 Due**
- **Sep 24 (Thu): Homework 2 Due**
- **Oct 8 (Thu): Homework 3 Due**
- **Oct 15 (Thu): Midterm Exam** (in class; closed book and closed notes)
- **Oct 29 (Thu): Homework 4 Due**
- **Nov 12 (Thu): Homework 5 Due**
- **Nov 24 and Nov 26 (Tue/Thu):** No class (Thanksgiving Break)
- **Dec 3 (Thu): Homework 6 Due (Non-graded)**
- **Dec 8 (Tue):** Last class meeting
- **Dec 9 (Wed):** Classes end
- **Dec 10 (Thu):** Reading Day (no class)
- **Dec 14 (Mon): Final Exam**, 7:45–9:45 a.m., Durham Hall 261 (closed book and closed notes)

Course Policies

Academic Honesty and the Graduate Honor Code

The tenets of the Virginia Tech Graduate Honor Code will be strictly enforced in this course, and all assignments will be subject to the stipulations of the Graduate Honor Code. Plagiarism, cheating (including inappropriate or unauthorized use of AI tools), falsification, academic sabotage, or other suspected violations of the Graduate Honor Code will be reported to the Graduate Honor System for review and adjudication. The Graduate Honor System is responsible for reviewing suspected violations and determining sanctions under its procedures. For more information, consult the current Graduate Honor System Constitution.

If you have questions or are unclear about what constitutes academic misconduct on an assignment, please speak with me. I take the Graduate Honor Code very seriously in this course.

You should attempt the problem sets on your own before consulting outside references. If you are stuck on a problem, you may discuss it with other students in the class, the TA, or me. This discussion should be limited to general ideas and strategies about the problem. Do not exchange any written material. You must cite any external references, such as books, websites, or individuals, that you use to improve your understanding. Failure to cite external references will be treated as cheating and will not be tolerated. Properly cited sources will not be penalized.

Generative Artificial Intelligence

Generative AI tools may be used to support learning in this course, including brainstorming, clarifying concepts, and debugging code, unless an assignment states otherwise. These tools may not be used during exams or to generate complete solutions, proofs, written explanations, or code that a student submits as their own. Any permitted AI assistance that contributes to submitted work must be disclosed in a brief note identifying the tool and explaining how it was used. Students remain responsible for verifying the correctness of all submitted work and properly citing external sources. Undisclosed or unauthorized use of generative AI will be treated as a potential violation of the Virginia Tech Graduate Honor Code. If you are uncertain whether a particular use is permitted, ask the instructor before proceeding.

Accommodations for Students with Disabilities

Virginia Tech welcomes students with disabilities into the University's educational programs. The University promotes efforts to provide equal access and a culture of inclusion without altering the essential elements of coursework. If you anticipate or experience academic barriers that may be due to a disability, please contact Services for Students with Disabilities (SSD) at 540-231-3788, by e-mail at ssd@vt.edu, or online at https://ssd.vt.edu/.

If you have an SSD accommodation letter, please meet with me privately as early in the semester as possible to discuss implementing your accommodations. Please provide reasonable notice, generally at least five business days and at least ten business days for final-exam accommodations.

Accessible Course Materials and Alternative Formats

Course materials are provided through Canvas and the course website. When a native HTML or Canvas version is available, it is the primary version intended for keyboard and screen-reader access; downloadable PDFs are provided for printing and offline reference. Canvas Ally may make alternative formats available for supported files. If any course page, equation, table, diagram, file, or activity presents an accessibility barrier, please contact the instructor and/or Services for Students with Disabilities so that an accessible alternative can be provided.

Inclusivity Statement

We understand that our members represent a rich variety of backgrounds and perspectives. Virginia Tech is committed to providing an atmosphere for learning that respects diversity. While working together to build this community, we ask all members to share their unique experiences, values and beliefs, be open to the views of others, honor the uniqueness of their colleagues, appreciate the opportunity that we have to learn from each other in this community, value each other's opinions and communicate in a respectful manner, keep confidential discussions that the community has of a personal (or professional) nature, use this opportunity together to discuss ways in which we can create an inclusive environment in this course and across the Virginia Tech community. The Virginia Tech Principles of Community are intended to increase access and inclusion and to create a community that nurtures learning and growth for all of its members. They are defined at https://www.inclusive.vt.edu/.

Course Evaluations

Students will be provided an opportunity to evaluate instruction in this course through Canvas using the University's standard procedures, which are administered by the Office of Institutional Effectiveness. The instructor may also provide additional informal surveys within the course to receive students' feedback and comments regarding the class. Feel free to provide your constructive feedback to the instructor regarding the class.

Tentative Course Outline

- **Introduction and Formulations (CH #1)**
  - LP Axioms/Standard Form
  - LP Formulations
  - Linear-integer modeling
- **LP Geometry (CH #2)**
  - Graphical Solution Method
  - Polyhedra/Extreme Points/Extreme Directions
  - Goldman Resolution Theorem
  - Linear Algebra Review (Matrix Inversion, Basic Feasible Solutions)
- **Optimization Software (Python + Gurobi)**
  - Introduction to Gurobi and Python-based modeling

  - Optimization Modeling and Solution Applications with `gurobipy`

- **Simplex Method (CH #3)**
  - Algebraic Version, Tableau Method
  - Optimality/Infeasibility/Unboundedness detection
  - Two-phase Method, Degeneracy and Cycling
  - Revised Simplex Method
- **Midterm Exam**
- **Sensitivity Analysis/Optimality Ranges (CH #5)**
- **Duality (CH #4)**
  - Formulating the Dual Problem
  - Weak and Strong Duality Theorems
  - Economic Interpretation
  - Complementary Slackness
  - Farkas' Lemma and Theorems of the Alternative
  - KKT Optimality Conditions
- **Variants of the Simplex Method (CH #4)**
  - Dual Simplex Method
  - Handling Simple Bounds
- **Advanced Topics (Time permitting) may include:**
  - Solving extremely high-dimensional LPs using sparse-aware or structure-exploiting interior point methods.
  - Recent breakthroughs in interior point methods that achieve faster runtimes than traditional combinatorial algorithms.
  - The Primal–Dual Hybrid Gradient (PDHG/PDLP) method for solving very large-scale LPs with applications in machine learning and network optimization.
  - Scalability and parallelization in first-order methods for LPs with billions of variables and constraints.
- **Final Exam**